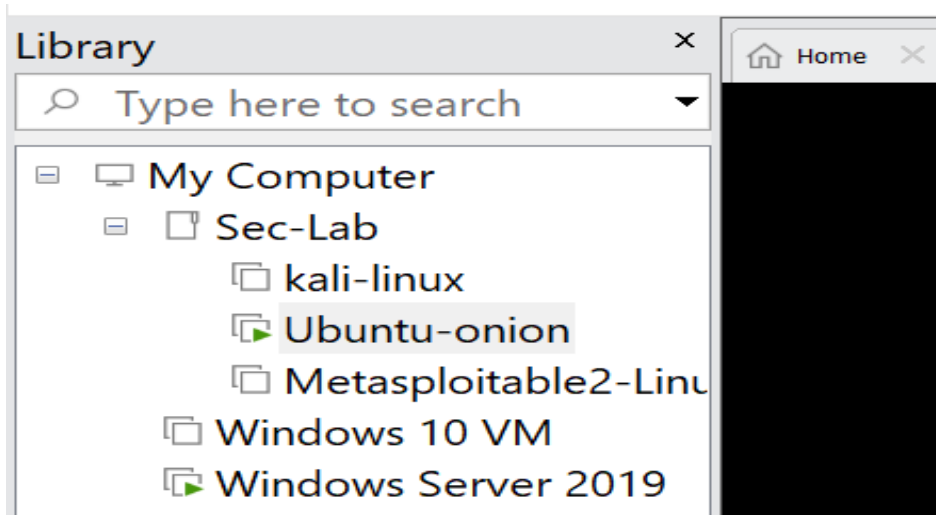


SOC Project 1:

Title

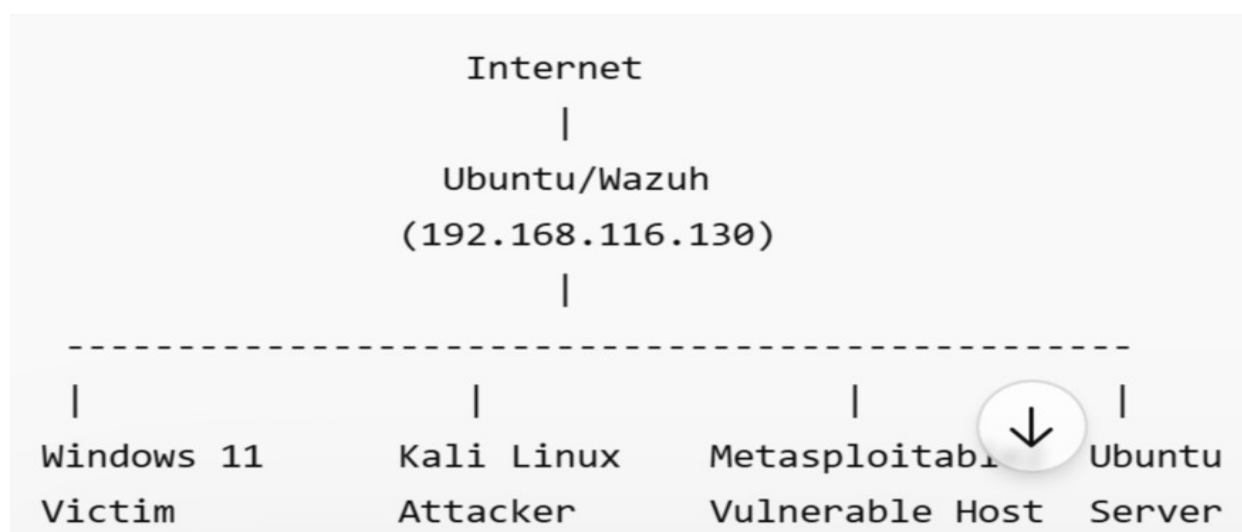
Brute Force Authentication Detection Using Wazuh SIEM

Project Ovetview



Lab Architecture

Figure 1



Environment

SIEM

Wazuh v4.14.5

Endpoint

RainhaStar
192.168.116.150
Windows Server 2019

Detection Scenario

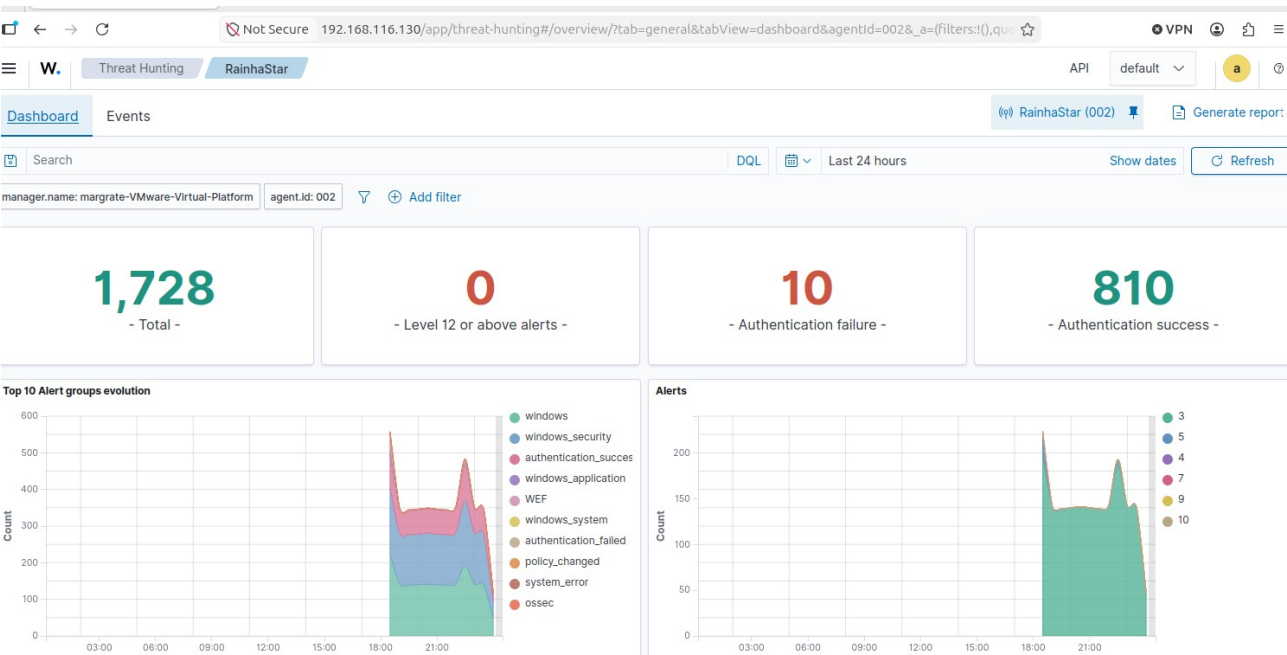
Wazuh detected multiple failed authentication attempts against the Windows Server Administrator account.

Wazuh Alert Dashboard

Figure 2

Alert

Multiple Windows Logon Failures



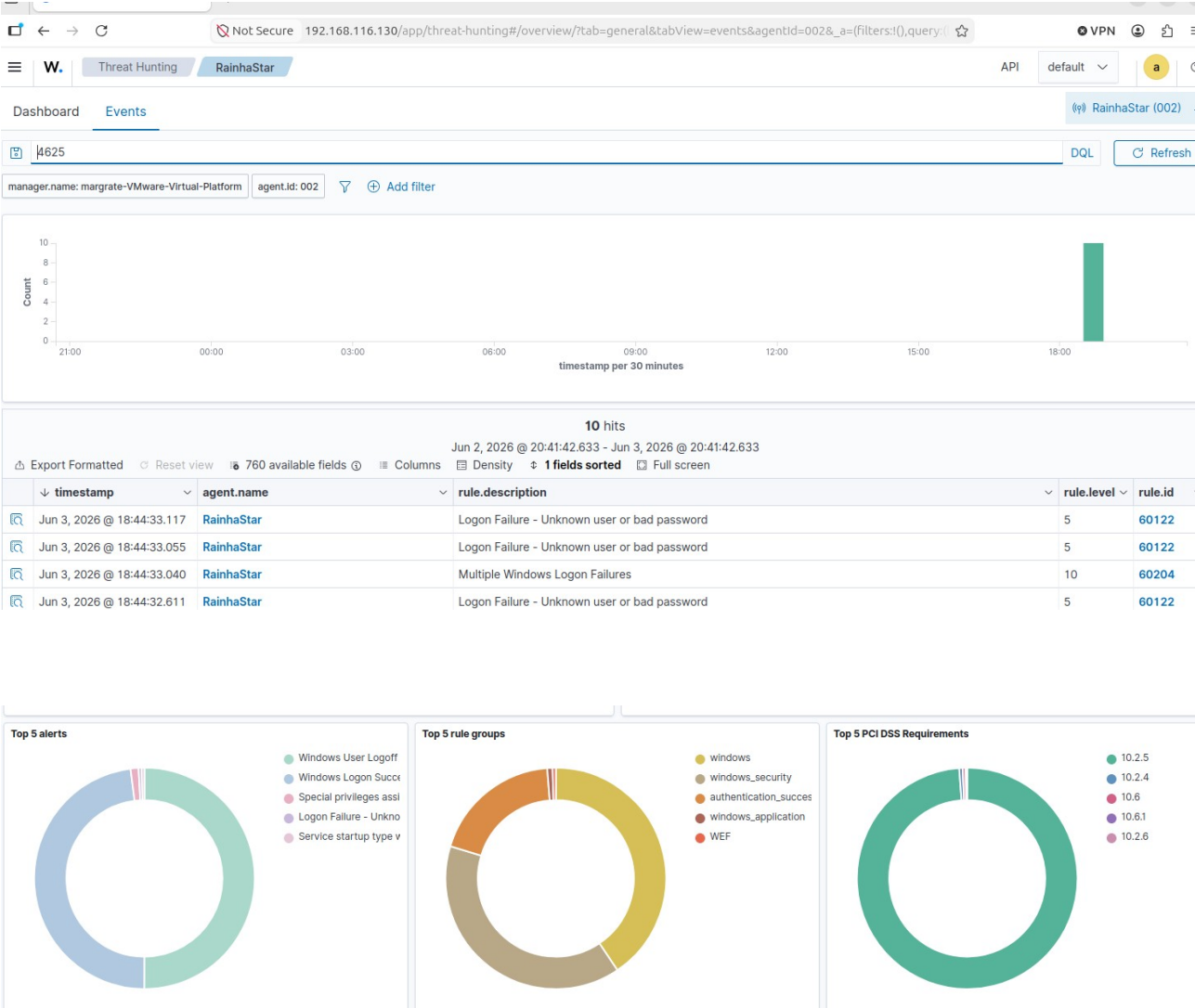
Rule ID

60204

Event Investigation (4625)

Figure 3

Severity



MITRE ATT&CK Mapping

Figure 4

MITRE ATT&CK Mapping

This activity aligns with



The screenshot shows the MITRE ATT&CK website interface. The top navigation bar is red with white text for 'MITRE | ATT&CK' and a search bar. Below the navigation bar, a sidebar on the left lists 'TECHNIQUES' with a scrollable list including 'Password Cracking', 'Password Spraying' (highlighted), 'Credential Stuffing', 'Credentials from Password Stores', and 'Exploitation for Credential Access'. The main content area is titled 'Brute Force: Password Spraying' and includes a dropdown for 'Other sub-techniques of Brute Force (4)'. The text describes the technique: 'Adversaries may use a single or small list of commonly used passwords against many different accounts to attempt to acquire valid account credentials. Password spraying uses one password (e.g. 'Password01'), or a small list of commonly used passwords, that may match the complexity policy of the domain. Logins are attempted with that password against many different accounts on a network to avoid account lockouts that would normally occur when brute forcing a single account with many passwords. [1]'. It also mentions that management services over commonly used ports are used when password spraying. On the right, a sidebar provides metadata: ID: T1110.003, Sub-technique of: T1110, Tactic: Credential Access, Platforms: Containers, ESXi, IaaS, Identity Provider, Linux, Network Devices, Office Suite, SaaS, Windows, macOS, Contributors: John Strand; Microsoft Threat Intelligence Center (MSTIC), and Version: 1.8.

T1110

Investigation Summary

A series of failed Windows logon attempts were generated against the Administrator account on the RainhaStar Windows Server 2019 endpoint.

The Wazuh agent collected Windows Security Event ID 4625 logs and forwarded them to the Wazuh Manager for analysis.

After correlating multiple failed authentication attempts, Wazuh generated Rule 60204 (Multiple Windows Logon Failures), indicating potential brute-force activity.

The activity was investigated through log analysis, event correlation, and MITRE ATT&CK mapping.

The investigation determined that the activity originated from authorized SOC lab testing. No successful authentication or unauthorized access occurred.

Analyst Findings

- Host: RainhaStar
- IP: 192.168.116.150
- Detection Source: Windows Security Logs

- Event ID: 4625
- Correlation Rule: 60204
- Result: Failed Authentication Attempts
- Impact: No successful compromise observed

Conclusion

Built a Wazuh SIEM home lab and investigated Windows authentication events, including Event ID 4625 failed logons and Rule 60204 brute-force detections. Performed threat hunting, event correlation, and incident documentation across Windows 11 and Windows Server 2019 endpoints.

